

Probability Refresher Notes

M08: Lecture 0 (Pages 1-8)

Callum Arnold

23 July, 2021

Contents

Random variables	1
CDFs and PDFs	2
Expectation	2
Example with exponential r.v	2
Conditional probability	3
Law of total probability	3
Law of total expectation	3
Law of total variance	4
Strong Law of Large Numbers	4
Central Limit Theorem	4

Random variables

- Random variable X is a function or variable whose value is generated by a random experiment e.g. coin flip.
- E.g. Binomial random variable
 - k successes in a fixed number of trials (n)
 - $X_i \sim \text{Bernoulli}(p)$
 - * $X \in \{0, 1\}$ with $\Pr(X = 1) = p, \Pr(X = 0) = 1 - p$ for $0 \leq p \leq 1$
 - The number of successes $S_n = \sum_{i=1}^n X_i$ is a *binomial r.v.*
 - $\Pr(S_n = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
- Negative binomial distribution
 - $X_1, X_2, \dots$, ordered Bernoulli(p)
 - Let $N = \min\{n : X_n = k\}$ be the number of trials until k successes
 - The first $n - 1$ trials must result in $k - 1$ successes, which can be modelled using a Binomial formula
 - * $\binom{n-1}{k-1} p^{k-1} (1 - p)^{(n-1)-(k-1)}$
 - n th trial must be a success with probability p

$$\begin{aligned}
\Pr(N = n) &= p \times \binom{n-1}{k-1} p^{k-1} (1-p)^{(n-1)-(k-1)} \\
&= \binom{n-1}{k-1} p^k (1-p)^{n-k} \\
&\text{for } n = k, k+1, \dots, \infty \\
\mu &= \frac{k}{p} \\
\sigma^2 &= \frac{k(1-p)}{p^2}
\end{aligned}$$

- Geometric distribution is a special type of Negative Binomial distribution
 - Number of trials until the first success occurs i.e. $k = 1$ for successes, or $n - r = 1$ for failures

$$\begin{aligned}
\Pr(N = n) &= \binom{n-1}{k-1} p^{k-1} (1-p)^{n-1} \\
&= p(1-p)^{n-1}
\end{aligned}$$

- A function that assigns probabilities to r.v values is called a **probability mass function**
 - Can be more generally defined using a **cumulative distribution function**

CDFs and PDFs

$$\begin{aligned}
F(x) &= \Pr(X \leq x) \\
f(x) &= \frac{d}{dx} F(x) \\
\int_a^b f(x) dx &= F(b) - F(a) = \Pr(a \leq X \leq b) \text{ for } a \leq b
\end{aligned}$$

Expectation

$$\begin{aligned}
E[g(X)] &= \int_{-\infty}^{\infty} g(x) dF(x) \\
&= \sum_{k=1}^{\infty} g(x_k) \Pr(X = x_k) \text{ for discrete r.v. } X \\
&= \int_{-\infty}^{\infty} g(x) f(x) dx \text{ for absolutely continuous r.v. } X
\end{aligned}$$

Example with exponential r.v

- Exponential r.v. has density $f(x) = \lambda e^{-\lambda x} 1_{\{x \geq 0\}}$
 - $\lambda > 0$ is the rate parameter
- Let $X \sim \text{Exp}(\lambda)$

$$\begin{aligned}
E(X) &= \int_0^{\infty} x \lambda e^{-\lambda x} dx \\
&= \left[\begin{array}{ll} f(x) = x & g'(x) = e^{-\lambda x} \\ f'(x) = 1 & g(x) = \frac{-e^{-\lambda x}}{\lambda} \end{array} \right] \\
&= \lambda \left[\cancel{xe^{-\lambda x}} + \frac{1}{\lambda} \int e^{-\lambda x} \right]_{-\infty}^{\infty} \\
&= \left[\frac{-e^{-\lambda x}}{\lambda} \right]_{-\infty}^{\infty} \\
&= \frac{1}{\lambda}
\end{aligned}$$

- Expectations are linear operations, so:
 - $E\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i E(X_i)$
 - If the random variables $X_1, \dots, X_n$ are independent:
 - * $\text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i)$

Conditional probability

$$\begin{aligned}
\Pr(B|A) &= \frac{\Pr(A \cap B)}{\Pr(A)} \\
E(X|A) &= \frac{E(X1_{\{A\}})}{\Pr(A)}
\end{aligned}$$

Discrete case

$$\Pr(X = x|Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)}$$

Continuous case

$$\begin{aligned}
F_{X|Y}(x|y) &= \frac{\int_{-\infty}^x f_{XY}(z, y) dz}{f_Y(y)} \\
f_{X|Y}(x|y) &= \frac{f_{XY}(x, y)}{f_Y(y)} \quad \frac{(\text{Joint density})}{(\text{Marginal density})} \\
\text{where: } f_Y(y) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dx
\end{aligned}$$

Law of total probability

If $B_1, \dots, B_n$ are mutually exclusive events

$$\begin{aligned}
\Pr(A) &= \sum_{i=1}^n \Pr(A \cap B_i) \\
&= \sum_{i=1}^n \Pr(A|B_i) \Pr(B_i)
\end{aligned}$$

Law of total expectation

$E(X)$ is scalar, but $E(X|Y)$ is a r.v.

For discrete r.v.s X, Y :

$$E(X|Y = y) = \sum_{k=1}^{\infty} x_k \Pr(X = x_k | Y = y)$$

$$\text{where: } E[E(X|Y)] = E[X]$$

Law of total variance

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 \\ &= E[E(X^2|Y)] - E[E(X|Y)]^2 \\ &= E[\text{Var}(X|Y) + E(X|Y)^2] - E[E(X|Y)]^2 \\ &= E[\text{Var}(X|Y)] + \{E[E(X|Y)^2] - E[E(X|Y)]^2\} \\ &= E[\text{Var}(X|Y)] + \text{Var}[E(X|Y)] \end{aligned}$$

Strong Law of Large Numbers

The empirical average of iid r.v.s converge to the theoretical expectation (average).

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n X_i = \mu$$

Central Limit Theorem

For large n , the empirical average behaves as $\mathcal{N}(\mu, \sigma^2/n)$, or, $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim \mathcal{N}(0, 1)$